
Predicting Secondary Educational Attainment from Global Poverty Rates

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1 Introduction

Secondary education is widely recognized as a cornerstone of human development and economic prosperity. Completing secondary school equips individuals with the skills and knowledge necessary to participate productively in labor markets and society. At the national level, high secondary completion rates are associated with stronger institutions, lower inequality, and sustained economic development. Yet despite decades of global policy attention, like the United Nations' Millennium Development Goals, significant disparities in secondary completion rates still exist across countries and regions. Understanding structural factors that drive these disparities is a critical challenge for researchers and policy makers alike.

Poverty is consistently identified in the literature as one of the most powerful barriers to educational attainment. At the household level, financial constraints limit the ability to bear the direct and indirect costs of schooling, like fees, materials, and foregoing child labor income. At the national level, countries with high rates of income poverty tend to have weaker educational infrastructure, lower public investment in schooling, and larger populations of children who are unable to access or sustain secondary education. However, while the inverse relationship between poverty and educational attainment is well established in principle, the empirical question of how well income poverty measures predict secondary completion rates at the cross-national level has room for further exploration. Much of the existing literature either focuses on individual or household level data, examines enrollment rather than completion, or does not systematically account for factors such as national income levels.

This study addresses that gap by examining whether income poverty headcount ratio predicts secondary school completion rates across countries, after controlling for GDP per capita. GDP per capita is included as a control because national income independently shapes countries' capacity to invest in education and students' opportunity cost of schooling [2]. By isolating the predictive contribution of the poverty headcount ratio net of this control, this study contributes to the literature in two important ways. First, it provides cross-national evidence on the poverty-secondary completion relationship using a standardized, widely available poverty measure, offering findings that are directly relevant to international development policy. Second, it highlights the extent to which GDP per capita and regional context mediate or moderate the role of poverty in shaping educational outcomes - an issue with significant implications for how poverty reduction and education policies are designed and targeted across different country contexts. This study uses multiple regression approaches to

quantify the relationship between education and poverty at the national level. This is accomplished by using country level data from the World Bank Group’s World Development Indicators database. Our primary dataset spans 2002–2022 across 63 countries.

2 Related Work

A gap exists in literature investigating whether poverty levels can predict educational outcomes in a global context while controlling for GDP. Existing research examining the relationship between poverty and education is often focused on the role of education in reducing poverty or predicting student dropout rates based on various factors in a specific country or region. Shi and Qamruzzaman [7] evaluated the role of education and financial inclusion in trying to reduce poverty in 68 lower-income countries between 1995-2018. They found that government investment in education positively assists poverty reduction, and increasing financial inclusion can decrease the poverty level.

The Quarterly Journal of Economics published an article [3] that introduces a distributional growth accounting framework to measure how much education has contributed to economic growth across income groups over the past several decades (1980-2019). It takes a micro database of household surveys from 95% of the world’s population in 2019 as a starting point and models what the world’s income would look like today if people were still only as educated as they were in 1980. When this imaginary ‘low-educated’ 2019 world is compared to the real 2019 world, the exact amount of how much global wealth came specifically from education can be calculated. It then tested this model on real-life scenarios where large education expansions took place and determined the model’s predictions were remarkably close to what occurred in the real world. The model demonstrates that education accounts for approximately 45% of global economic growth and 60% of income gains for the world’s poorest 20%.

Psyridou et al. [5] utilizes parameters like student academic skills, cognitive skills, and behavior to predict upper secondary education dropout. It looks at a 13-year longitudinal dataset based on student information in Finland and uses machine learning classification methods like Balanced Random Forest and Bagging Decision Tree. It used metrics of 6-fold cross-validation, confusion matrices, accuracy, precision, recall, and AUC score. Their most effective classifier ended up being the Balanced Random Forest.

Hassan et al. [4] predicted student dropout rates from primary or secondary school for students in Somaliland, an independent African country. Indicators from the data used for prediction included grade, age, school type, household income, and type of housing. The supervised machine learning algorithms used were logistic regression, probit regression, naive bayes, decision tree, random forest, support vector machine, and K-nearest neighbors. Some metrics used to assess model performance included accuracy, sensitivity, and AUC. Their top performing models were the decision tree and SVM, both getting an accuracy of 0.988, and random forest getting an AUC result of 0.988. It was concluded that grade and age were crucial in predicting dropout rates, and future models should be built suited to those.

Rodríguez et al. [6] developed a model for the Chilean educational system to predict school dropout with longitudinal data (since 2004) obtained by the Ministry of Education. Their best performing method was the use of decision trees in with gradient boosting ensembles, Light Gradient Boosting Machine (LightGBM). Their metrics for binary classification included recall, precision, sensitivity, specificity, GM score, and F1 score. One of their indicators for predicting dropout was socioeconomic status, which includes household income and parental schooling. Other indicators included variables at individual, family, school, and extra-scholar levels.

Delprato and Farieta [1] utilized multilevel modeling to investigate determinants of educational transitions in sub-Saharan Africa. The three-level structure includes individuals nested within communities, which are nested within countries. Random intercept logistic regression and random slope models were applied to control for heterogeneity across communities/countries and analyze if the impact of household wealth varied randomly across different units. They found that education coverage has grown significantly over the past 50 years. However, the variation is much higher between communities within the same country than between countries themselves, suggesting targeted community level interventions may be more effective.

While these findings are valuable in examining how educational improvements can reduce poverty and finding the root causes of dropout rates in specific areas, this paper aims to take a more global approach to investigating how poverty levels predict educational outcomes. It will also analyze the proportion to which GDP per capita mediates this relationship. This paper employs linear regression analysis to test whether poverty headcount ratios predict secondary completion rates globally, after accounting for GDP per capita.

3 Methods

3.1 Data Sourcing and Preliminary Decisions

For this project, we used the World Bank’s World Development Indicators database [8]. WBG is a collection of smaller organizations focused on ending poverty through stabilizing governments and communities. The organization’s robust data collection and education focused approach form the backbone of this project. Per WBG privacy and use documents, data is collected through “mostly official means”, interpreted to mean that governments agree to send them data they have collected themselves. Other data collection methods include obtainment from other non profits and from household surveys conducted by WBG themselves. The data is stored in an open access online database, easily filterable by country, timeframe, and marker.

Our team then chose indicators (both by examining the database and referencing previous work) in order to predict secondary education completion rates from poverty rates and GDP among all countries around the world. After selecting the indicators, we created a large, downloadable dataset for 266 countries with each of the seven indicators, making a 1,867 x 54 size dataset. We selected three indicators for our target variable of secondary completion: percent of the population over 25 years of age who have at least completed upper secondary school separated into female, male, and total populations. We then selected two indicators for our predictor variable of poverty: poverty headcount ratio (given as a percentage) for those living under \$8.30 a day and the poverty headcount ratio for those living under their own country’s national poverty line. Lastly, two indicators for the other predictor variable of GDP were selected: GDP per capita in US\$ adjusted for inflation in 2015, and GDP per capita percent of annual growth. It was intentional to include more indicators than necessary so that after data cleaning we could further investigate which ones we would want to use. In further analysis, the feature set was expanded to include Gini index, political stability percentile rank, urban population percentage, and secondary education expenditure as a percentage of GDP per capita, sourced from the same World Bank database. This expanded dataset enabled analysis of additional structural predictors beyond poverty and GDP.

3.2 Data Preprocessing and Exploratory Analysis

After loading the data, the data cleaning and preparation phase began. Multi-indices were used to sort the data by country first then series; now only 266 country rows were present in the dataset. The names for each indicator in the series column were simplified, columns were reorganized, trailing rows were dropped, and the placeholders ‘..’ from original dataset were replaced with “NaN”. Next we wanted to analyze our missing data so as to clean it, and a heatmap was made to visualize this. We decided to establish a range of years from our dataset to use moving forward, which was 2002 through 2022. This range seemed to have the least missing data in the heatmap, with the year 2016 containing the most amount of data. ¹ We then filtered the data to use only three indicators of interest, ‘ed_attain_total’ (total percent of population who completed upper secondary), ‘poverty_ratio_wage’ (poverty headcount ratio living under \$8.30 a day), and ‘GDP_const’ (GDP per capita in US\$ adjusted). We chose to use the educational attainment total with hopes of looking into gender breakdowns in future investigation; we chose the poverty at \$8.30 a day because we had more data for this poverty indicator across countries than the other; and we chose the constant GDP because it gives a more stable and general way to control for GDP than annual growth. We then dropped all countries with more than 50% missing values and made a duplicate dataframe with all numeric values for plotting and machine learning purposes. Geopandas was used to plot a world map highlighting the kept countries. We also looked at bar charts for poverty ratios, one sorted by the most recent year and one by the year with the most data.

Overall, it was observed that there was a lot more missing data than expected with a limited number of countries to be used for analysis. In addition, our previous method removed countries based on a

missing data threshold across any of the series, resulting in an imbalance of missing data between markers. To mitigate this, we went back and kept countries with less than 60% of missing data across any given series, which resulted in keeping 63 countries in our dataframe. 2 We then did more data visualization with a Geopandas world map, and many more countries were highlighted blue on the map. 3 Similar bar charts were made to view poverty ratios per country as well as GDP, 4 and a choropleth map showed GDP constant and GDP growth in side by side shaded maps of the world for our selected countries between 2002 and 2022 5.

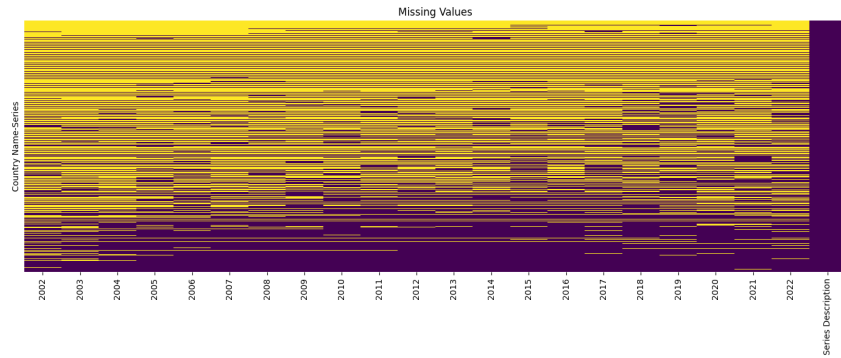


Figure 1: Spread of missing values (yellow) prior to cleaning

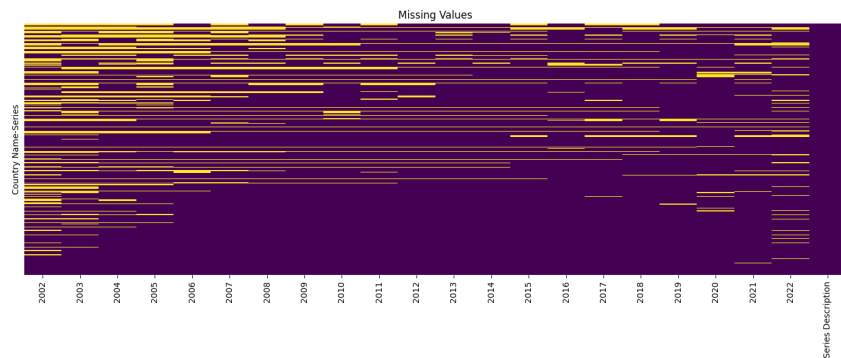


Figure 2: Missing values (yellow) post cleaning and filtering

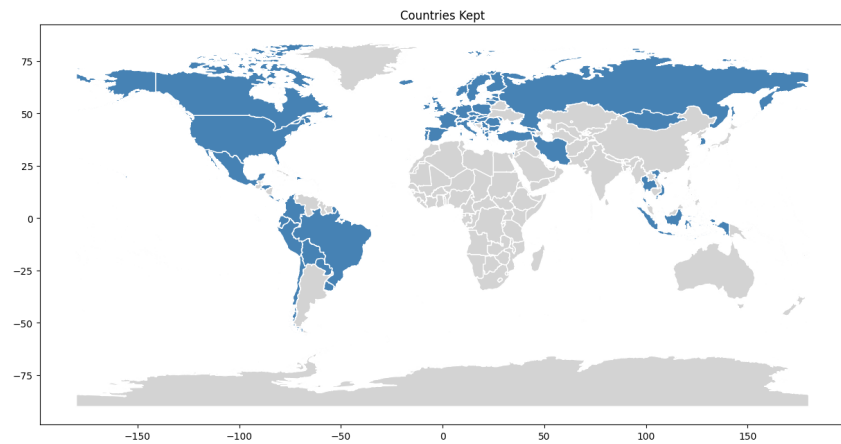


Figure 3: Map of countries kept in our dataset

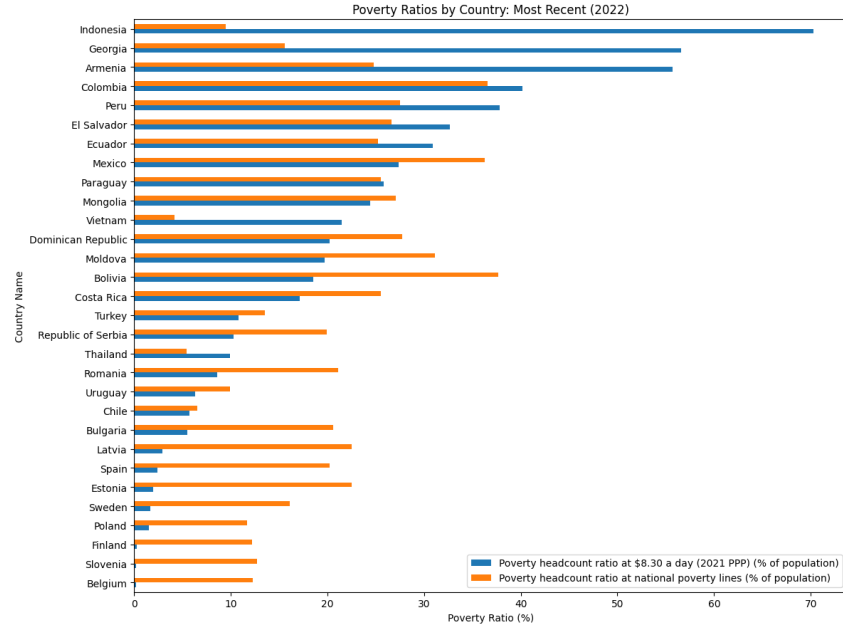


Figure 4: Showcasing the divergence of poverty ratio metrics

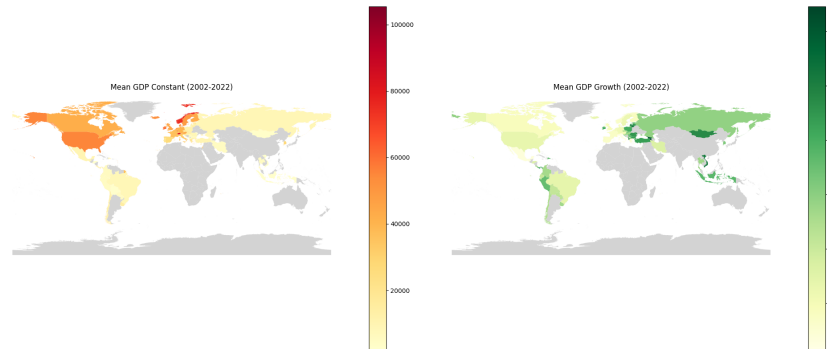


Figure 5: Map of GDP and GDP growth

After cleaning the dataset, determining a threshold for when to drop countries from our analysis due to excessive missing values and imputing the remaining gaps, we had a clearer picture of what would be available for our analysis. After dropping data below the 40% threshold we were left with 63 countries. Geographically, retained countries are concentrated in the Americas and Eurasia with noteworthy gaps in Africa, South Asia, Australia and Greenland. This geographic clustering may limit how generalizable our model is to these underrepresented areas of the world. The poverty levels for the countries we have in our cleaned data show substantial variation, indicating a diverse economic context for our analysis and reflecting countries at varying stages of economic development. Interestingly, the two poverty metrics (poverty headcount ratio at \$8.30 a day and poverty headcount ratio at national poverty lines) diverge considerably. This indicates that national poverty lines vary by country and that poverty metrics are sensitive to the choice of threshold. Similarly, mean GDP levels vary widely across our sample. To compensate for this, we chose to scale our GDP per capita constant US\$ variable via Logarithmic Transformation. This diversity in poverty and GDP levels strengthens our analysis by providing adequate variation in key variables.

The geographic distribution of mean GDP per capita shows a distinct pattern across regions. Higher income countries tend to be concentrated in North America and Western Europe, while lower income are concentrated in Latin America and select regions of Eurasia. This geographic concentration of

higher GDP levels mirrors the distribution of our retained sample and may influence regional patterns in our analysis.

To handle missing data we chose to linearly interpolate in accordance with Gethin [3].

$$y = y_1 + \frac{(x - x_1)(y_2 - y_1)}{x_2 - x_1} \quad (1)$$

Linear interpolation is appropriate here because educational attainment, poverty rates, and GDP measures tend to change gradually and monotonically over time within a given country, making a linear approximation between known values reasonable. However, if there were any sudden spikes in the data, such as those that would be present due to sudden economic or social instability, they would be lost to the interpolation method. The nature of this method's dependence on existing data both before (y_1) and after (y_2) the interpolated value y leaves gaps on either end of the sequence of data. We chose to fill the tails of the data with forward fill and backward fill, which simply duplicate the the nearest known point. We made a new dataframe where all NaN were replaced. The formerly unknown data points and their interpolated values can be visualized in a sample plot for a given country 7 .

By linearly imputing the data we have filled in the gaps with plausible values, allowing us to maximize the data we do have and minimize the bias that would be introduced by missing information. This approach preserves the temporal trends in our data while ensuring we can conduct regression analysis with complete observations across all countries and years in our sample.

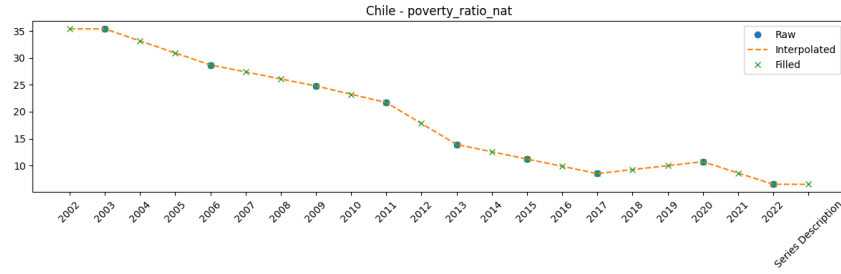


Figure 6: Imputation example

3.3 Predictive Model Selection

Our primary predictive method was Ordinary Least Squares (OLS) Regression. OLS minimizes the sum of squared residuals between observed and predicted values, estimating the linear relationship between a continuous dependent variable and one or more independent predictors. It assumes a linear relationship, independence of observations, and normally distributed residuals. This method was chosen due to its continuous outcome prediction and its ability to control for multiple features simultaneously, directly addressing our research question. Our first modeling attempt began with predicting educational attainment from poverty at \$8.30 per day variable and log GDP. We then expanded our methods to OLS on other features 1.

Educational_Attainment_Male ~ Poverty_Ratio_Wage + log_GDP_Constant
Educational_Attainment_Total ~ Poverty_Ratio_National + log_GDP_Constant
Educational_Attainment_Male ~ Poverty_Ratio_National + log_GDP_Constant

Table 1: Other features and targets modeled with OLS

We modeled OLS with an interaction term; adding a poverty $\times$ log GDP interaction to capture whether the poverty-education relationship varies by economic development level. In addition, we also tried OLS with a Logarithmic Transform on the target educational attainment. Other linear models used include Ridge Regression and a Fixed Effects Model controlling for time-invariant country-level differences. Non-linear methods Random Forest and XGBoost were attempted. Group K-Fold cross

validation was used on both models to evaluate generalizability, splitting the data such that no country appeared in both training and testing folds simultaneously.

Finally, we attempted our original OLS Regression on poverty and log GDP, this time with an 80/20 training testing split. This model performed the best out of the methods we attempted, and is the primary focus of our results.

4 Results

4.1 Ordinary Least Squares Regression

After data cleaning and preprocessing was completed, we moved on to creating an ordinary least squares (OLS) linear regression model. We first utilized the StatsBase package in Python to train a model on all of the available clean data. This initial model achieved an R-squared of 0.234, indicating that the two predictor variables, poverty and log GDP per capita, were explaining about 23.4% of the variance in the outcome variable, secondary educational attainment. Figure 7 shows the resulting model summary. Adjusted R-squared is nearly identical to the unadjusted R-squared, confirming that the two predictor variables are meaningfully contributing to the model without overfitting the data. Similarly, the model has an F-statistic of 201.7 and Significance F, shown as Prob(F-statistic), of 3.59e-77 implying that predictors are likely not causing the variance by chance and that the model is statistically significant and predicts better than chance.

OLS Regression Results						
Dep. Variable:	Educational_Attainment_Total	R-squared:	0.234			
Model:	OLS	Adj. R-squared:	0.233			
Method:	Least Squares	F-statistic:	201.7			
Date:	Fri, 10 Apr 2026	Prob (F-statistic):	3.59e-77			
Time:	16:58:48	Log-Likelihood:	-5761.2			
No. Observations:	1323	AIC:	1.153e+04			
Df Residuals:	1320	BIC:	1.154e+04			
Df Model:	2					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	33.4734	9.056	3.696	0.000	15.707	51.239
Poverty_Ratio_Wage	-0.3165	0.042	-7.624	0.000	-0.398	-0.235
log_GDP_Constant	3.5483	0.881	4.026	0.000	1.819	5.277
Omnibus:	5.467	Durbin-Watson:	0.124			
Prob(Omnibus):	0.065	Jarque-Bera (JB):	5.366			
Skew:	0.151	Prob(JB):	0.0684			
Kurtosis:	3.075	Cond. No.	527.			

Figure 7: Original OLS Regression with poverty and log GDP targeting educational attainment

Since the goal of the study was to actually predict educational attainment by using poverty rate and GDP, we created a second model using the Scikit-learn package with a train/test split of 80/20. After training the model on 80% of randomly selected data points, the resulting model achieved an R-squared of 0.267 and Mean Squared Error of 330.018 on the test set. The residuals of the test set are normally distributed, indicating that the model's errors follow a normal distribution, with a mean of zero, confirming that the model has captured the main data patterns 8.

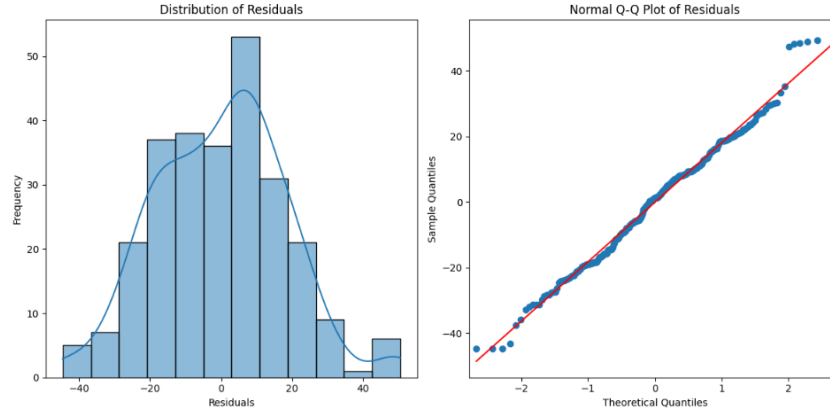


Figure 8: 80/20 Training/Testing Split Residuals

Lastly, we created multiple tertiary models in order to compare with the results of the first two models. We created models using other predictors and outcome variables, for example using national poverty rates and log GDP per capita to predict the female and male secondary educational completion rates separately (as opposed to the total secondary educational completion rate). All of these tertiary models proved to have lower R-squared measures, confirming that the predictors in the initial models, poverty ratio at \$8.30/day and log GDP per capita, were the best of the indicators that we found from the World Bank Group’s Development Indicators. Although the R-squared value was not as high as we initially thought, the models do confirm that there is a quantifiable impact of poverty and GDP on secondary educational attainment.

4.2 Further Analysis and Future Work

In addition to the models explored above, we created an OLS regression model with an expanded feature set to include Gini index, political stability, urban population percentage, secondary education expenditure, and year as additional predictors. This expanded OLS model achieved an R-squared of 0.433, suggesting that inequality and governance measures capture additional variance. Gini index was the strongest individual predictor, indicating that income inequality is more predictive of secondary educational attainment than poverty headcount ratio after controlling for other factors. Future work may include models with the Gini index as a primary predictor, as well as examine the relationship between education, poverty, and gender.

5 Conclusion

Secondary education plays a crucial role in a country’s development, both for economic growth and societal well-being. However, young people drop out of education before reaching the end of their school terms, and we wanted to investigate how poverty and GDP could be reasons for this. The World Bank Group provided us with a large dataset of 266 countries, and there were challenges that went along with that. A significant amount of data cleaning was required, such as inconsistent country names, metadata to be removed, and series titles to be shortened. Another major issue we faced right away was missing data. Dozens of countries had missing data that we needed to train our models on, so we had to make some decisions; we first selected the years 2002-2022 because they had more data. Then we selected a threshold of removing countries with 50% of more missing data across all series, and this left us with 174 countries but still too many missing values. In our second round after that, we removed all countries with 60% of more missing data from *any* series, which left us with 63 countries and much better data to work with.

The variables we chose for our models were total secondary educational attainment as the target, and GDP constant and poverty at \$8.30 a day as the predictors. It was found that log GDP had more of a clear positive correlation with educational attainment, while poverty showed more noise but in general a negative correlation. Regarding the models, ordinary least squares (OLS) performed the best, with an R-squared of 0.234. This concluded that the predictor variables were able to explain

about 23.4% of the variance in educational attainment. This shows that these economic factors of GDP and poverty contribute to education, but there are still several factors that go into predicting education completion. The OLS did get an F-statistic of 201.7, indicating statistically significant results. Comparably, a second Linear Regression model with sklearn and an 80/20 test split resulted in an R-squared of 0.267.

We investigated using several other modeling techniques with this data; the original OLS performed the best though there were some metric trade-offs. While we explored interaction terms and regularization methods, those approaches did not substantially improve predictive power, indicating the moderate explanatory capacity is attributed to the intrinsic complexity of predicting educational outcomes rather than overfitting. Adding an interaction term of Poverty Ratio x GDP to our regression model gave an R-squared of 0.263, however, it also produced a very high condition number indicating multicollinearity. Other models like Random Forest and XGBoost were tested, and while the Random Forest model produced an R-squared of 0.635, the mean R-squared from cross-validation was -0.09, suggesting the model was significantly overfitted. Our original OLS regression led to the strongest results with the variables we were using; simpler models proving to be more reliable.

As far as future directions in this line of study, there are clearly many other factors that can go into predicting educational attainment. Our exploratory analysis incorporating the Gini index, political stability, urban population percentage, and education expenditure improved model performance substantially with an R-squared of 0.433, appearing to account for much more of the variance in the educational outcome data. However, this expanded model exhibited multicollinearity concerns, highlighting a trade-off between explanatory power and coefficient interpretation that future research should address. Future studies could clearly get improved results and stronger causal insights by incorporating additional variables into predictive models. Another future investigation mentioned earlier was educational attainment split by gender. It's very possible that different factors go into predicting secondary education completion for different genders.

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